

Knowledge Discovery in Text Corpora with Computational Linguistics and Information Theory

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Abstract

For a human to learn a new advanced scientific concept from scratch, the learning path will include materials of increasingly difficult and specialized character. This work establishes a framework based on concepts from computer linguistics and information theory to capture this linguistic complexity with an entropy-based complexity measure for texts, which depends on how typical a document's vocabulary is relative to the general corpus. A cross-entropy based measure is further introduced that simultaneously quantifies document similarity and indicates the direction of complexity increase. This asymmetric measure is then used to find geodesics between simple, introductory documents and complex, advanced ones, potentially uncovering useful progressions of prerequisite documents. The methods are validated qualitatively on an exemplary corpus of machine learning and mathematics related documents.

1 Introduction

As science progresses and new knowledge is generated, scientific language evolves alongside it. Wholly new words emerge, and existing words are used in novel ways to express new ideas. In many fields, the work being done at the absolute frontier of science may be understood only by a handful of people, a fact reflected in the highly specialized terminology that is inaccessible to most readers foreign to the field.

Scientific progress is, however, often incremental. Even the most novel ideas build on previously existing ideas and literature, and therefore also on previously existing language. In principle, most advanced ideas should be understandable by any determined individual given sufficient time, even without prior knowledge in the field.

Pedagogically, the progression of materials consumed during such a learning journey follows a somewhat predictable trajectory, well exemplified by mathematics and physics education: it proceeds from the most basic concepts through increasingly advanced material, into university-level content (perhaps bridged by popular science texts), and finally to specialized research at the frontier. Along this path, the language of the materials becomes progressively more complex and less accessible to a general reader.

The goal of this work is to create a mathematical framework that captures this complexity, or "advancedness," of a document using tools from computer linguistics and information theory. The ultimate aim is to model learning paths as geodesics along an entropy gradient.

The core building blocks of the framework are as follows. Term Frequency-Inverse Document Frequency (TF-IDF) [2] treats the inverse document frequency of a word as a measure of its information content: words that appear across many documents carry little discriminative information, while rare words are highly informative. Latent Semantic Analysis (LSA) [3] via Singular Value Decomposition (SVD) of the TF-IDF vectors uncovers the latent semantic structure of the corpus, and a Non-negative SVD (NN-SVD) [4] transformation allows document representations to be interpreted as probability distributions over semantic components. This connects document complexity directly to compressibility: documents using common, widely-shared vocabulary are well-represented by the dominant SVD components (low entropy), while documents with specialized vocabulary require many components for accurate reconstruction (high entropy). Finally, cross-entropy between document distributions serves as a directed similarity measure related to the Kullback-Leibler divergence: it is easy to encode a specialized document using the vocabulary of a general one, but not vice versa. This asymmetry defines a directed kernel from which geodesics can be computed, tracing natural learning progressions through the corpus.

2 Data

2.1 Dataset Description

The dataset used in this study is derived from the OpenWeb-Math [5] corpus, specifically filtered and curated for mathematical machine learning content. The data collection and preprocessing pipeline involved several stages:

Source Selection: Documents were extracted from the OpenWeb-Math dataset with URL filtering restricted to sources including "wikipedia.org", "github.io", and "nature.com".

Topic-based Retrieval: Zero-shot BERTopic [6] was employed to retrieve documents matching specific mathematical and machine learning topics, including linear algebra, statistics, probability theory, information theory, machine learning, deep learning, statistical learning, reinforcement learning, and neural networks.

Preprocessing Pipeline: Retrieved documents underwent standard NLP preprocessing techniques including tokenization, stopword removal, and stemming to prepare the text for subsequent analysis.

Each document contains text content, URL, and topic classification, with topics ranging from basic concepts like "Introducing matrices" to advanced topics like "LSTM networks".

2.2 Data Properties

This dataset provides an ideal testbed for evaluating semantic relationships and learning pathways in mathematical and machine learning concepts. The hierarchical nature of mathematical knowledge, from foundational linear algebra to advanced neural architectures, allows us to validate whether our geodesics capture meaningful semantic progressions. The documents vary significantly in complexity and abstraction level, providing a natural gradient for testing our entropy-based characterization methods.

3 Methods

3.1 Text Preprocessing and Vectorization

Our pipeline begins with text preprocessing and Frequency-Inverse Document Frequency (TF-IDF) [2] vectorization:

$$\mathbf{X}_{t,d} = \text{TF}(t, d) \cdot \log \left(\frac{N}{|\{d' : t \in d'\}|} \right) \quad (1)$$

where t represents a term, d a document, and N the total number of documents. The preprocessing includes removal of terms appearing in more than 50% of documents and terms appearing in fewer than 5 documents, along with Gensim preprocessing for text cleaning.

3.2 Singular Value Decomposition

To manage computational complexity while preserving semantic information, we apply SVD:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (2)$$

where $\mathbf{X} \in \mathbb{R}^{n \times m}$ is the TF-IDF matrix, and we retain the top $k = 5000$ components. This decomposition is also known as Latent Semantic Analysis [3], and provides dimensionality reduction from vocabulary size to manageable dimensions, noise reduction through low-rank approximation, and computational efficiency for subsequent operations.

3.3 Non-negative SVD (NN-SVD) Transformation

Since diffusion processes require non-negative values, we apply a Non-negative SVD (NN-SVD) [4] transformation that preserves the dominant singular vector components while ensuring non-negativity. This transformation also serves as an effective initialization strategy for Non-negative Matrix Factorization (NMF).

The NN-SVD approach addresses the fundamental challenge that standard SVD produces factors with both positive and negative entries, while many applications (including diffusion processes and NMF) require non-negative representations. For each singular vector component, we preserve the orientation that captures the majority of the signal energy:

$$\mathbf{U}_{\text{nonneg}}[:, i] = \begin{cases} \max(\mathbf{U}[:, i], \epsilon) & \text{if } \|\mathbf{U}_+[:, i]\|_2 \geq \|\mathbf{U}_-[:, i]\|_2 \\ \max(-\mathbf{U}[:, i], \epsilon) & \text{otherwise} \end{cases} \quad (3)$$

where $\mathbf{U}_+$ and $\mathbf{U}_-$ denote positive and negative components respectively, and $\epsilon = 10^{-5}$ prevents numerical issues.

This transformation strategy offers several advantages. Energy preservation is achieved by selecting the orientation with larger L_2 norm, retaining the maximum amount of information from each singular vector while satisfying non-negativity constraints. The approach maintains the semantic structure captured by SVD while enabling probabilistic interpretations required for diffusion analysis.

3.4 Entropy-based Similarity Kernel

The NN-SVD matrix fundamentally serves as a change of basis for the original TF-IDF matrix, projecting the document representations from the high-dimensional vocabulary space into a lower-dimensional semantic space with non-negative coordinates.

The NN-SVD provides a new basis where each document is represented as:

$$\mathbf{X}_{\text{nonneg}} = \mathbf{U}_{\text{nonneg}}^T \mathbf{X} \quad (4)$$

where $\mathbf{U}_{\text{nonneg}}$ forms the new basis vectors derived from the non-negative transformation of the left singular vectors.

We normalize the transformed non-negative matrix to form probability distributions:

$$\mathbf{P} = \frac{\mathbf{X}_{\text{nonneg}}}{\sum_i \mathbf{X}_{\text{nonneg}}[i, :]} \quad (5)$$

and compute a cross-entropy based kernel:

$$\mathbf{K}_{ij} = - \sum_k \mathbf{P}_{ik} \log_2(\mathbf{P}_{jk}) \quad (6)$$

which can be written compactly as a matrix product

$$\mathbf{K} = -\mathbf{P} (\log_2 \mathbf{P})^T \quad (7)$$

where $\log_2$ is applied element-wise to $\mathbf{P}$.

3.4.1 SVD Entropy as Document Generality and Complexity Measure

The SVD entropy provides a novel characterization of document generality through the fundamental principle that documents using common, widely-shared vocabulary can be more efficiently compressed using the dominant SVD components, while documents with specialized vocabulary require more components for accurate representation.

$$H_{\text{SVD}}(d) = \frac{\mathbf{K}_{dd}}{\log_2(n)} \quad (8)$$

This measure captures document characteristics along multiple dimensions. Documents with low SVD entropy exhibit high generality, extensively using vocabulary patterns captured by the top SVD components. These documents employ common terminologies and conceptual frameworks found across the corpus, typically representing introductory material, overviews, or foundational concepts that can be efficiently "explained" using the dominant semantic patterns in the corpus. Examples include basic tutorials, general introductions, and widely-applicable principles.

Conversely, documents with high SVD entropy demonstrate high specialization, requiring many SVD components for accurate reconstruction. These documents use terminology and conceptual patterns that are rare or unique within the corpus, representing advanced, specialized, or highly technical content that cannot be efficiently compressed using only the dominant corpus-wide patterns. Examples include research papers, technical specifications, and domain-specific advanced discussions.

The compression-based intuition reveals that SVD entropy directly measures how "compressible" a document is using the corpus-wide semantic patterns. Just as simple images with common features compress well with few JPEG coefficients while complex

images with unique details require many coefficients for faithful reproduction, general documents using common vocabulary patterns compress well with few SVD components (low entropy) while specialized documents with unique terminology require many SVD components (high entropy).

This measure has important pedagogical implications, automatically identifying documents by their position in the learning hierarchy. Low entropy documents often represent accessible entry points, while high entropy documents represent advanced material requiring substantial background knowledge. The SVD entropy thus serves as an automatic "difficulty level" indicator based on vocabulary specialization rather than subjective assessments.

This compression-based complexity measure provides an objective, corpus-relative assessment of document generality that scales naturally with the semantic diversity of the collection.

3.4.2 Asymmetric Cross-Entropy Kernel Interpretation

The cross-entropy kernel $\mathbf{K}_{ij}$ has a profound information-theoretic interpretation that captures the fundamental asymmetries in semantic relationships between documents. Specifically, $\mathbf{K}_{ij}$ measures the expected number of bits required to encode document i using the vocabulary distribution of document j as the coding scheme.

This asymmetric formulation reveals several important semantic phenomena. Information flow and teaching relationships emerge when $\mathbf{K}_{ij} < \mathbf{K}_{ji}$, indicating that document j provides a more efficient encoding for document i than vice versa. This suggests that j contains the "vocabulary foundation" or conceptual framework needed to understand i , and in educational contexts, this indicates that j could serve as a prerequisite or explanatory text for i .

Generalization versus specialization asymmetries become apparent when document j is a comprehensive overview and document i is specialized. In such cases, $\mathbf{K}_{ij}$ (encoding specialist using generalist's vocabulary) will typically be lower than $\mathbf{K}_{ji}$ (encoding generalist using specialist's vocabulary). This reflects the natural relationship where general texts provide broad vocabulary that can describe specialized concepts, but specialized texts lack the breadth to efficiently describe general concepts.

The asymmetry also enables conceptual dependency detection, capturing which documents can "explain" others through shared conceptual vocabulary. Documents with lower cross-entropy when used as encoders likely represent foundational or prerequisite knowledge. Additionally, documents that serve as bridges between domains will exhibit asymmetric relationships with texts from both domains, potentially identifying interdisciplinary or survey documents.

This information-theoretic perspective transforms document similarity from a symmetric geometric problem into an asymmetric knowledge flow analysis, revealing the directional relationships inherent in learning pathways and conceptual dependencies.

3.5 Diffusion Process Construction

We construct a similarity matrix using a Gaussian kernel applied to the entropy-based divergences:

$$\mathbf{S}_{ij} = \exp\left(-\frac{\mathbf{K}_{ij}}{\sigma^2}\right) \quad (9)$$

where $\sigma = 1.5$ is a bandwidth parameter. The similarity matrix is then row-normalized:

$$\mathbf{M}_{ij} = \frac{\mathbf{S}_{ij}}{\sum_k \mathbf{S}_{ik}} \quad (10)$$

forming a column-stochastic Markov transition matrix.

3.6 Eigendecomposition and Diffusion Distance

We compute the eigendecomposition of the transpose of the Markov matrix:

$$\mathbf{M}^T \mathbf{v}_i = \lambda_i \mathbf{v}_i \quad (11)$$

With this eigendecomposition, the simulated diffusion (random walks) of t steps on $\mathbf{M}$ can be efficiently computed:

$$\mathbf{M}^t = \mathbf{V} \text{diag}(\lambda_1^t, \dots, \lambda_k^t) \mathbf{V}^{-1} \quad (12)$$

For non-integer values of t and negative eigenvalues, the complex valued root is calculated and only the real part of that is kept.

The diffusion distance [1] between points i and j at time t is defined as:

$$D_t(i, j) = \sqrt{\sum_k \frac{(\mathbf{M}^t[i, k] - \mathbf{M}^t[j, k])^2}{\pi_k}} \quad (13)$$

where π_k is the stationary distribution of the Markov matrix $\mathbf{M}$.

3.7 Geodesic Path Finding

We construct a k -nearest neighbors graph using diffusion distances and apply Dijkstra's algorithm [7] to find the shortest paths:

Algorithm 1 Geodesic Path Finding

Input: Diffusion distances $\mathbf{D}_t$, parameter k

Output: Shortest path between source and target

for each document i **do**

 Find k nearest neighbors based on $\mathbf{D}_t[i, :]$

 Add edges to KNN graph with diffusion distances as weights

end for

Apply Dijkstra's algorithm for shortest path computation

4 Results

4.1 Geodesic Path Analysis

We computed a geodesic path between two conceptually distant documents: a foundational linear algebra review (entropy 0.190) at one end and a Wikipedia article on sparse coding (entropy 0.726) at the other. Sparse coding, which describes the representation of stimuli by the strong activation of a small subset of neurons, sits at the intersection

of information theory, computational neuroscience, and deep learning, making it a natural high-entropy endpoint in this corpus. The path was found by running Dijkstra’s algorithm on a k-nearest-neighbor graph weighted by diffusion distances, as described in the Methods section. The result is a sequence of 13 intermediate documents that forms a curriculum-like progression, automatically discovered from the geometry of the corpus without any manual annotation.

4.2 Visualization Analysis

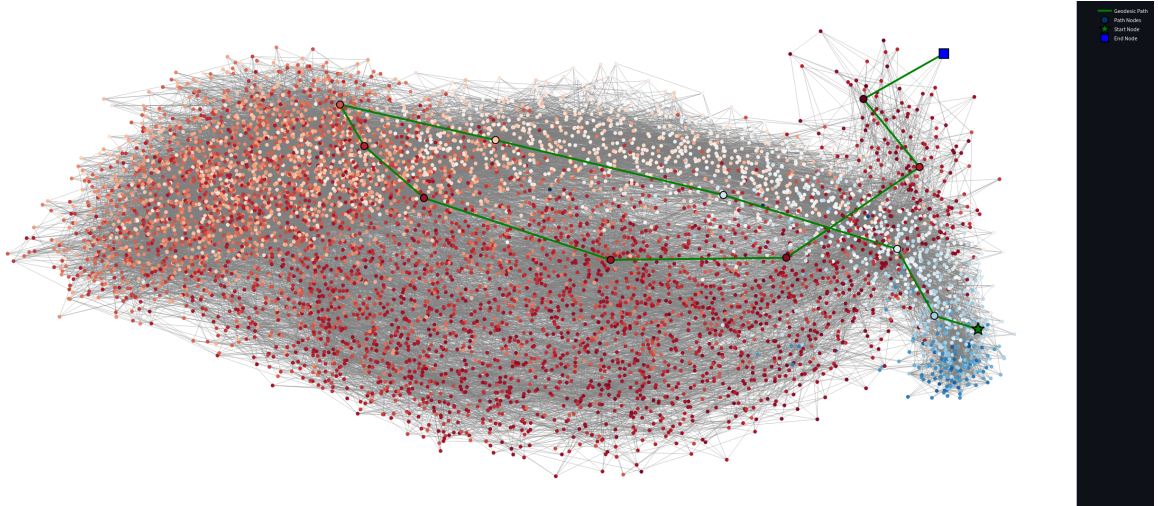


Figure 1: KNN graph of the document corpus visualized with a spring layout [8], colored by SVD entropy (blue: low entropy, red: high entropy). The green star marks the start of the geodesic (low-entropy linear algebra review in the blue cluster), the blue square marks the end (the sparse coding article, entropy 0.726). Intermediate path nodes are shown as hollow circles connected by the green geodesic line.

Figure 1 shows the KNN graph embedded in 2D using a spring (force-directed) layout [8]. The node color encodes SVD entropy: blue nodes have low entropy (general, foundational documents) and red nodes have high entropy (specialized, advanced documents). The structural separation is immediately visible: the bottom-right cluster is almost entirely blue, corresponding to introductory mathematical content such as linear algebra tutorials and probability primers. The large left and central regions are predominantly red, representing advanced and specialized material. This spatial separation confirms that the diffusion geometry naturally organizes documents along a complexity gradient without any supervision.

The geodesic (green line) begins at the green star in the bottom-right blue cluster and ends at the blue square near the upper-right boundary of the graph. The path first traverses within the low-entropy cluster, then bridges toward the central and upper regions through several intermediate documents. Notably, the path does not cut directly through the densest red region; instead it follows the periphery, suggesting that the algorithm identifies a chain of intermediate documents that share sufficient vocabulary overlap to allow efficient encoding, rather than making large jumps across conceptually distant material.

4.2.1 Geodesic Path Progression Analysis

Table 1: Geodesic Path: 13-Step Progression from Foundational Linear Algebra to Specialised Advanced Topics

Step	Document	Normalized Entropy
0	<i>Review: Linear Algebra</i> — Matrix notation, transpose, multiplication as column linear combinations, invertibility.	0.190
1	<i>Linear Algebra for Fun and Profit</i> — Matrices via social networks, correlation matrices, image compression, Fibonacci dynamics.	0.371
2	<i>Edge Detection: The Sobel Operator</i> — Convolution kernels for edge detection; linear algebra meets signal processing.	0.432
3	<i>Linear Algebra Workbook: Matrix Operations</i> — Strang Ch. 1.3; vector spaces; 3Blue1Brown <i>Essence of Linear Algebra</i> .	0.408
4	<i>Linear Algebra Workbook: Three Perspectives</i> — Strang Ch. 2.1; elimination; three views of matrix–vector multiplication.	0.535
5	<i>Linear Algebra Workbook: LU Decomposition</i> — Strang Ch. 2.6; LU factorisation and practical linear system solving.	0.637
6	<i>Extreme Value Threshold Selection (threshr)</i> — R package for i.i.d. threshold diagnostics; transition to applied statistics.	0.673
7	<i>Bayesian Statistics Dissertation</i> — Prior/posterior distributions, decision-making under uncertainty, probabilistic modelling.	0.694
8	<i>DADS: Dynamics-Aware Discovery of Skills</i> — Unsupervised skill discovery for model-based RL; MuJoCo locomotion planning.	0.696
9	<i>Lean Community: Sum is Linear</i> — Formal verification of sum linearity in the Lean theorem prover.	0.708
10	<i>ConfigSpace User Guide</i> — Hyperparameter spaces: conditional parameters, forbidden clauses, AutoML sampling.	0.681
11	<i>Diagnosing Heart Diseases with Deep Neural Networks</i> — CNNs for echocardiogram classification; advanced applied deep learning.	0.735
12	<i>Sparse Coding</i> — Neural codes via sparse population activation; links information theory, neuroscience, and deep learning.	0.726

Table 1 lists the 13 documents along the discovered path together with their SVD entropy values. The entropy rises from 0.190 at the start to values above 0.70 at the end, with only minor non-monotonicities reflecting short local detours through the graph.

The path opens with two closely related introductory linear algebra documents. The first is a comprehensive linear algebra appendix (fncbook.github.io) covering the core vocabulary of the field: matrix dimensions, the transpose, matrix multiplication as linear combinations of columns, inner and outer products, and the connection between invertibility and uniqueness of solutions to linear systems. Its SVD entropy is very low (0.190),

reflecting that this vocabulary is extremely common across the corpus and compresses well onto the dominant singular vectors. The second document, a slide-based introduction titled *Linear Algebra for Fun and Profit* (sheaves.github.io, entropy 0.371), covers similar definitional ground but motivates matrices through broader applications including social network analysis, statistical correlation matrices, image compression, and discrete dynamical systems such as the Fibonacci recurrence.

The path then moves into applied linear algebra territory, and here a particularly striking pattern emerges. Steps 3, 4, and 5 are three successive workbook assignments from a single course on linear algebra by Theron Hitchman, structured around Gilbert Strang’s textbook *Introduction to Linear Algebra*. Crucially, the geodesic recovers these assignments in their correct pedagogical order: Chapter 1.3 (matrix operations and vector spaces, entropy 0.408), followed by Chapter 2.1 (three perspectives on matrix–vector multiplication and Gaussian elimination, entropy 0.535), followed by Chapter 2.6 (LU factorisation and its role in practical linear system solving, entropy 0.637). The algorithm had no access to chapter numbers, course structure, or any metadata; it recovered the correct textbook ordering purely from the diffusion geometry of the vocabulary distributions. Between step 2 and step 3, the path also passes through an edge-detection document applying convolution kernels to images, which shares vocabulary with both basic linear algebra and applied numerical computation, providing a natural conceptual bridge. The entropy rises from 0.43 to 0.64 across this segment as the vocabulary shifts from broadly shared matrix terminology toward more specialised algorithmic and numerical content.

Steps 6 onward cross into statistical and machine learning domains. An R package reference for extreme-value threshold selection (entropy 0.673) marks the transition from linear algebra into applied statistics; a Bayesian statistics dissertation (0.694) introduces probabilistic modelling vocabulary; and a review of the DADS skill-discovery paper (0.696) brings in reinforcement learning terminology. Later steps cover formal theorem proving in Lean (0.708), AutoML hyperparameter configuration spaces (0.681), and convolutional neural networks applied to medical imaging (0.735). The path terminates at a Wikipedia article on sparse coding (0.726), which characterises neural representations through sparse population activation and ties together vocabulary from information theory, neuroscience, and deep learning, making it a semantically rich and appropriately specialised endpoint.

The overall entropy progression, with only small local reversals, demonstrates that the diffusion geodesic successfully recovers a pedagogically meaningful ordering from the raw geometric structure of the corpus, without any explicit difficulty labels or curriculum design.

5 Discussion

5.1 Methodological Advantages

Our diffusion-based methodology offers several significant advantages over traditional text mining approaches. By treating the text corpus as a manifold with intrinsic geometry, we enable natural distance computations that reflect semantic relationships. The diffusion process naturally incorporates a time parameter, allowing analysis at multiple scales and the discovery of hierarchical relationships. The use of entropy-based similarity measures and diffusion distances ensures that the intrinsic semantic relationships are preserved

throughout the analysis. Optimizations using Numba and efficient data structures enable the processing of medium-scale document collections with reasonable computational resources. The geodesic paths and diffusion map embeddings provide results that are not only effective for analysis but also interpretable in terms of semantic relationships and document complexity.

5.2 Limitations and Future Work

While our methodology demonstrates promising results, several limitations warrant consideration. The current implementation is suitable for medium-scale corpora, but scaling to very large document collections would require distributed computing approaches or approximate methods for eigendecomposition and distance computations. The method involves several parameters (SVD components, diffusion time, kernel bandwidth) that may require tuning for different domains and corpus characteristics. Additionally, the approach has been validated primarily on mathematical and machine learning content, and evaluation on other domains would strengthen the generalizability claims.

Future work could explore adaptive parameter selection, distributed implementations for large-scale corpora, and validation across diverse text domains to establish broader applicability.

6 Conclusion

This work presents a novel framework for text mining that leverages diffusion geometry and information theory to reveal the underlying semantic structure of document corpora. Our approach successfully combines TF-IDF vectorization, SVD-based dimensionality reduction, entropy-based similarity measures, and diffusion processes to create a comprehensive analysis pipeline.

The key innovations include the development of SVD entropy as a measure of document generality and complexity, the use of asymmetric cross-entropy kernels to capture directional semantic relationships, and the computation of semantic geodesics that represent meaningful learning pathways between concepts.

Evaluation on a mathematical machine learning corpus demonstrates the method’s effectiveness in discovering semantic relationships, characterizing document complexity, and finding interpretable paths between conceptually distant topics. The approach provides both quantitative measures through diffusion distances and qualitative insights through geodesic path analysis.

The framework opens new possibilities for educational technology, where semantic geodesics could guide personalized learning paths, and for knowledge discovery, where the geometric structure of document collections can reveal hidden relationships and progression patterns in specialized domains.

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